

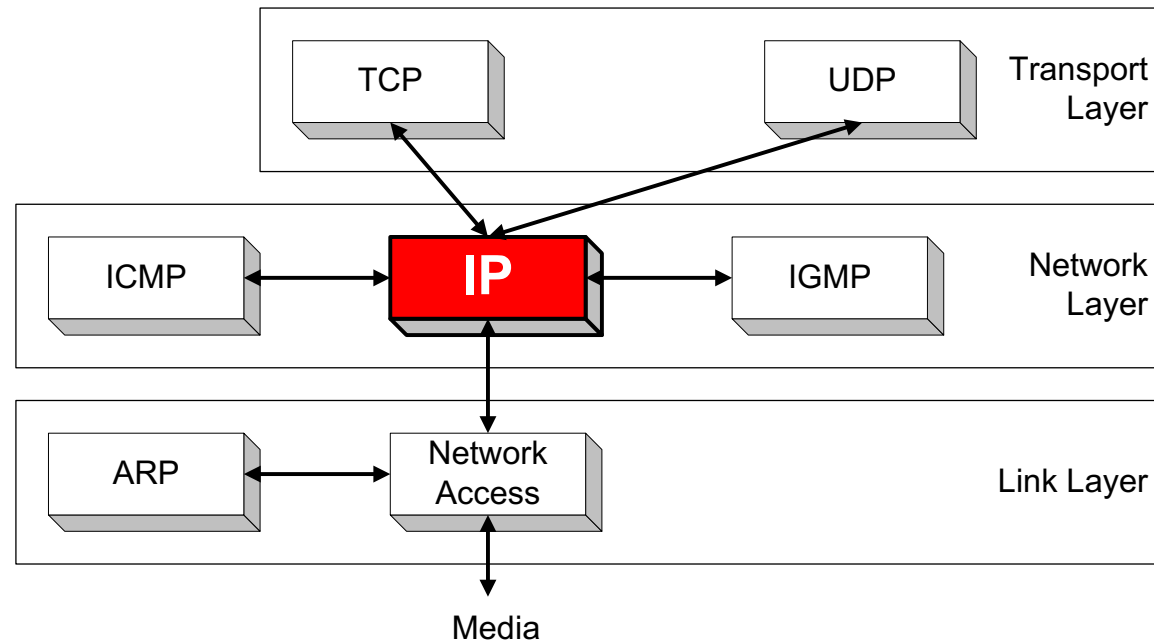
# IP - The Internet Protocol

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- ▶ Introduction
- ▶ IP Addressing
- ▶ Subnetting
- ▶ Variable Length Subnet Mask (VLSM)

# Orientation

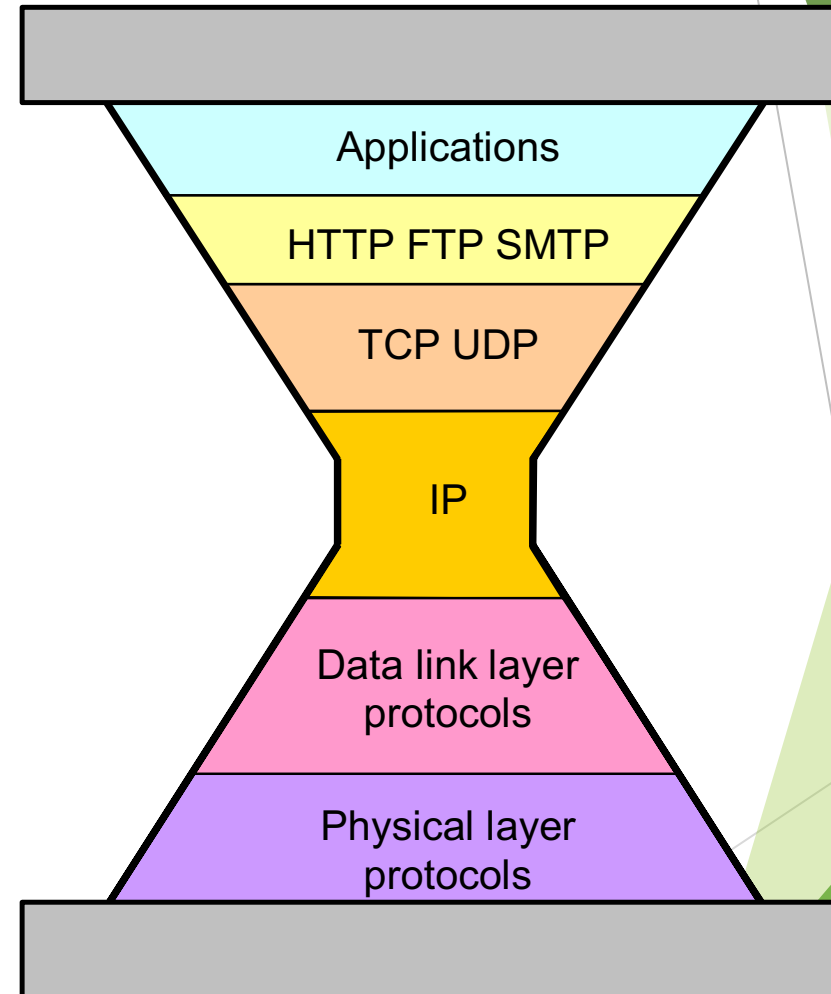
- ▶ IP (Internet Protocol) is a Network Layer Protocol.



- ▶ IP's current version is Version 6 (IPv6).

# IP: The waist of the hourglass

- ▶ **IP is the waist of the hourglass of the Internet protocol architecture**
- ▶ Multiple higher-layer protocols
- ▶ Multiple lower-layer protocols
- ▶ Only one protocol at the network layer.



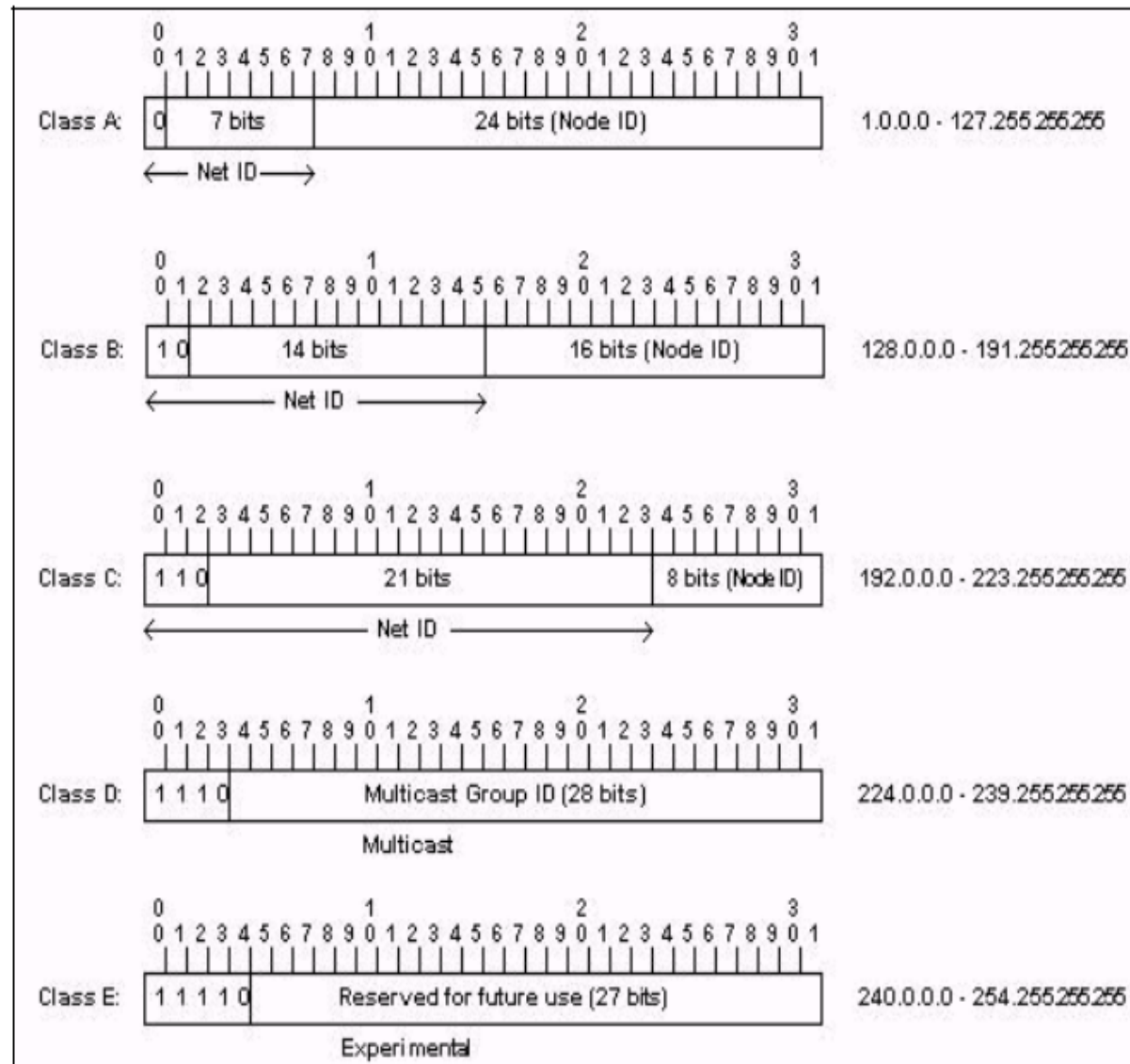
# IP Service

- ▶ Delivery service of IP is minimal
- ▶ IP provide provides an **unreliable connectionless** best effort service (also called: “datagram service”).
  - ▶ **Unreliable:** IP does not make an attempt to recover lost packets
  - ▶ **Connectionless:** Each packet (“datagram”) is handled independently. IP is not aware that packets between hosts may be sent in a logical sequence
  - ▶ **Best effort:** IP does not make guarantees on the service (no throughput guarantee, no delay guarantee,...)
- ▶ Consequences:
  - ▶ Higher layer protocols have to deal with losses or with duplicate packets
  - ▶ Packets may be delivered out-of-sequence

# IP Addresses

- An IP address is an address used to uniquely identify a device on an IP network.
- The address is made up of 32 binary bits which can be divisible into a network portion and host portion with the help of a subnet mask.
- 32 binary bits are broken into four octets (1 octet = 8 bits)
- Dotted decimal format (for example, 172.16.81.100)

# IP Address Classes (1)



# IP Address Classes (2)

- ▶ Class A: The first octet is the network portion. Octets 2, 3, and 4 are for subnets/hosts
- ▶ Class B: The first two octets are the network portion. Octets 3 and 4 are for subnets/hosts
- ▶ Class C: The first three octets are the network portion. Octet 4 is for subnets/hosts

# Private Address Range

| Address Class | Reserved Address Space        |
|---------------|-------------------------------|
| Class A       | 10.0.0.0 - 10.255.255.255     |
| Class B       | 172.16.0.0 - 172.31.255.255   |
| Class C       | 192.168.0.0 - 192.168.255.255 |



# Network/Subnet Masks

- ▶ Distinguishes which portion of the address identifies the network and which portion of the address identifies the node.
- ▶ Default masks:
  - Class A: 255.0.0.0/8
  - Class B: 255.255.0.0/16
  - Class C: 255.255.255.0/24

# Subnetting

- ▶ Creates multiple logical networks that exist within a single Class A, B, or C network.
- ▶ If you do not subnet, you will only be able to use one network from your Class A, B, or C network, which is unrealistic
- ▶ Each data link on a network must have a unique network ID, with every node on that link being a member of the same network

Example



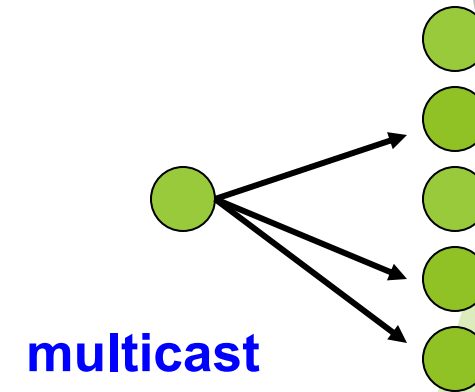
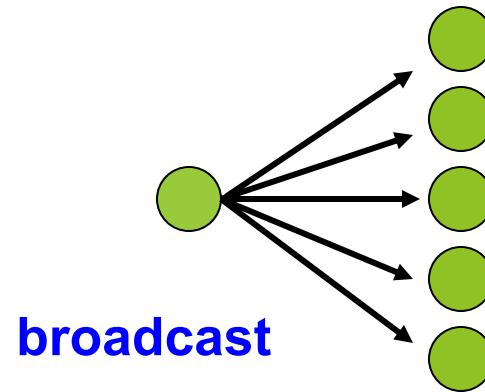
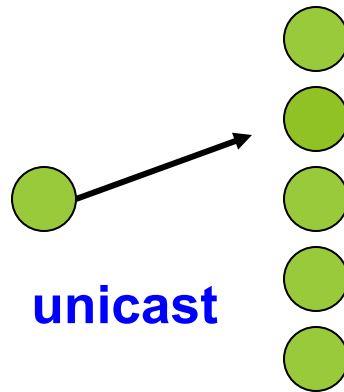
# Benefits of Subnetting

- Reduced network traffic
- Optimized network performance
- Simplified management
- Facilitated spanning of large geographical distances

# IP Service

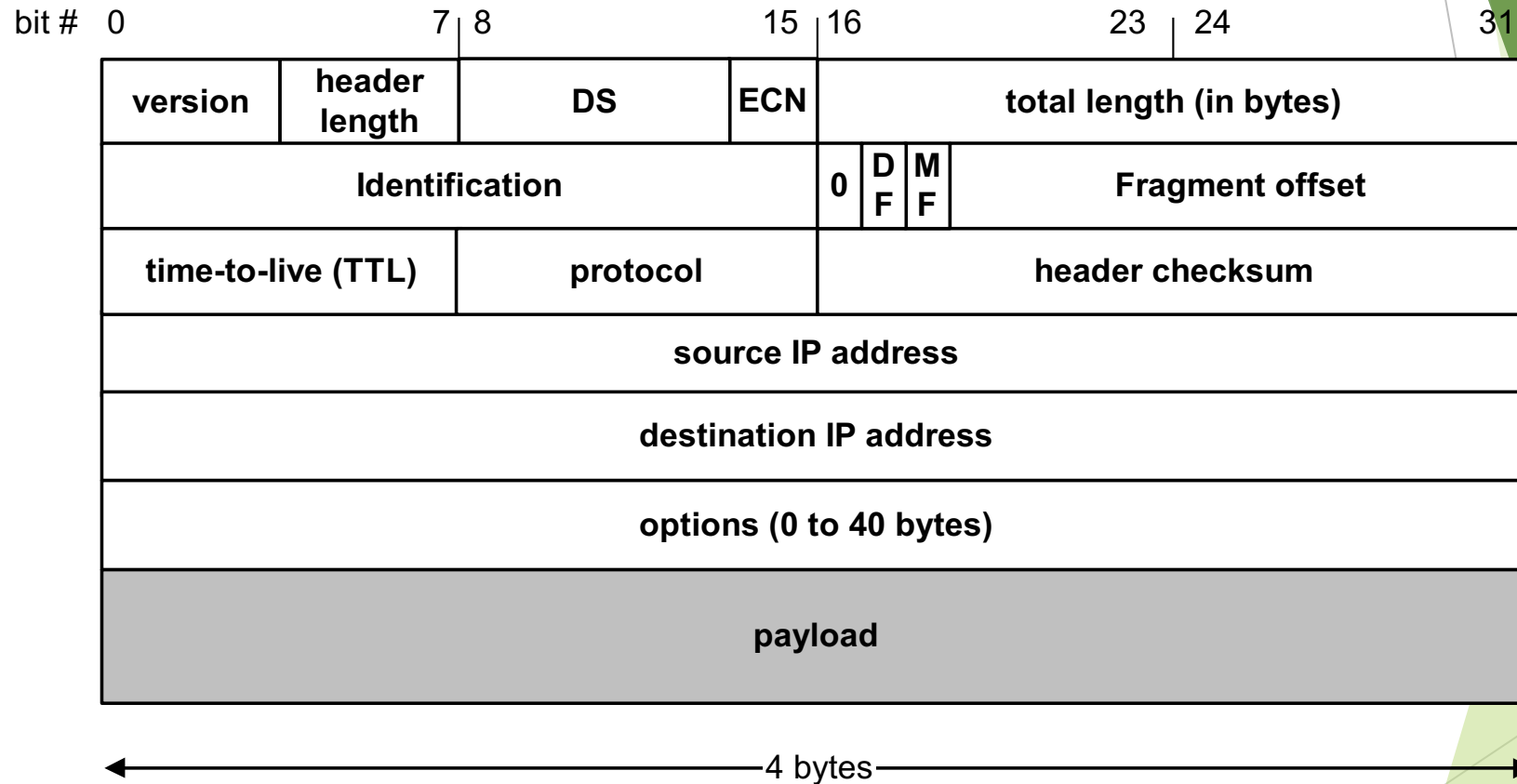
- ▶ IP supports the following services:

- ▶ one-to-one (unicast)
- ▶ one-to-all (broadcast)
- ▶ one-to-several. (multicast)



- ▶ IP multicast requires support of other protocols (IGMP, multicast routing)

# IP Datagram Format



- ▶ 20 bytes ≤ **Header Size** <  $2^4 \times 4 \text{ bytes} = 60 \text{ bytes}$
- ▶ 20 bytes ≤ **Total Length** <  $2^{16} \text{ bytes} = 65536 \text{ bytes}$

# Fields of the IP Header (1)

- ▶ **Version (4 bits):** current version is 4, next version will be 6.
- ▶ **Header length (4 bits):** length of IP header, in multiples of 4 bytes
- ▶ **DS/ECN field (1 byte)**
  - ▶ This field was previously called as Type-of-Service (TOS) field. The role of this field has been re-defined, but is “backwards compatible” to TOS interpretation
  - ▶ **Differentiated Service (DS) (6 bits):**
    - ▶ Used to specify service level (currently not supported in the Internet)
  - ▶ **Explicit Congestion Notification (ECN) (2 bits):**
    - ▶ New feedback mechanism used by TCP

# Fields of the IP Header (2)

- ▶ **Identification (16 bits):** Unique identification of a datagram from a host. Incremented whenever a datagram is transmitted
- ▶ **Flags (3 bits):**
  - ▶ First bit always set to 0
  - ▶ DF bit (Do not fragment)
  - ▶ MF bit (More fragments)

Will be explained later → Fragmentation

# Fields of the IP Header (3)

## ► Time To Live (TTL) (1 byte):

- Specifies longest paths before datagram is dropped
- Role of TTL field: Ensure that packet is eventually dropped when a routing loop occurs

Used as follows:

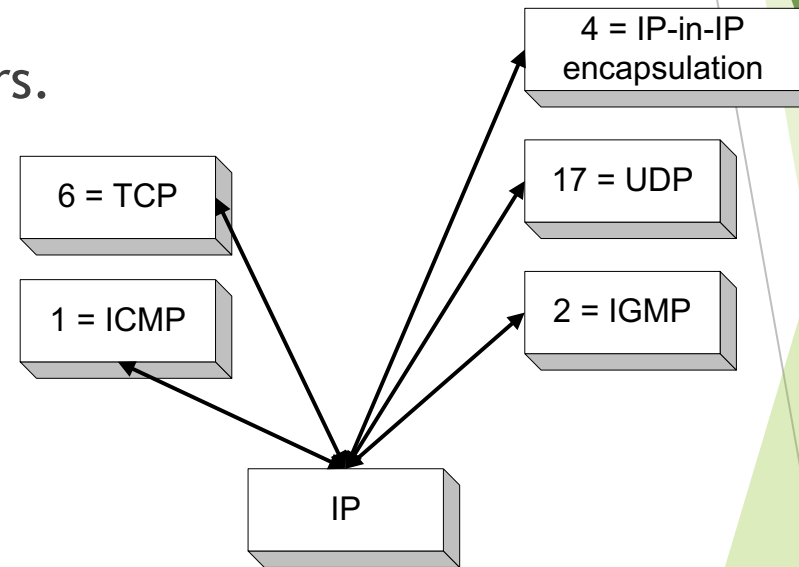
- Sender sets the value (e.g., 64)
- Each router decrements the value by 1
- When the value reaches 0, the datagram is dropped



# Fields of the IP Header (4)

## ► Protocol (1 byte):

- Specifies the higher-layer protocol.
- Used for demultiplexing to higher layers.



- **Header checksum (2 bytes):** A simple 16-bit long checksum which is computed for the header of the datagram.

# Fields of the IP Header (5)

## ► Options:

- Security restrictions
- Record Route: each router that processes the packet adds its IP address to the header.
- Timestamp: each router that processes the packet adds its IP address and time to the header.
- (loose) Source Routing: specifies a list of routers that must be traversed.
- (strict) Source Routing: specifies a list of the only routers that can be traversed.

- **Padding:** Padding bytes are added to ensure that header ends on a 4-byte boundary

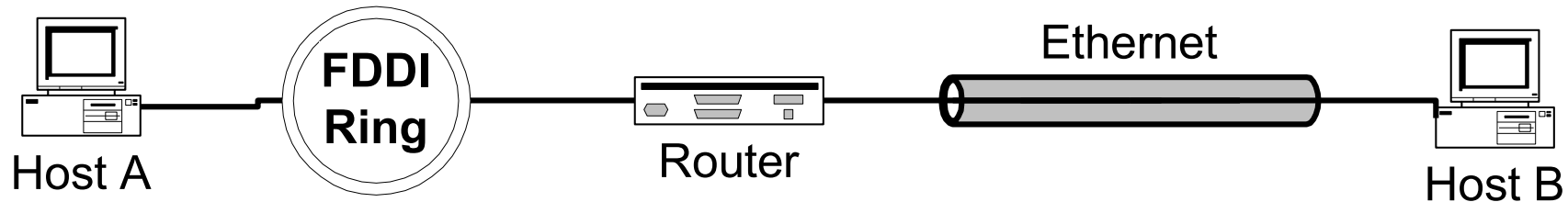
# Maximum Transmission Unit (MTU)

- ▶ Maximum size of IP datagram is 65535, but the data link layer protocol generally imposes a limit that is much smaller
- ▶ Example:
  - ▶ Ethernet frames have a maximum payload of 1500 bytes
    - IP datagrams encapsulated in Ethernet frame cannot be longer than 1500 bytes
- ▶ The limit on the maximum IP datagram size, imposed by the data link protocol is called **maximum transmission unit (MTU)**
- MTUs for various data link protocols:

|           |      |           |            |
|-----------|------|-----------|------------|
| Ethernet: | 1500 | FDDI:     | 4352       |
| 802.3:    | 1492 | ATM AAL5: | 9180       |
| 802.5:    | 4464 | PPP:      | negotiated |

# IP Fragmentation

- What if the size of an IP datagram exceeds the MTU?  
IP datagram is fragmented into smaller units.
- What if the route contains networks with different MTUs?

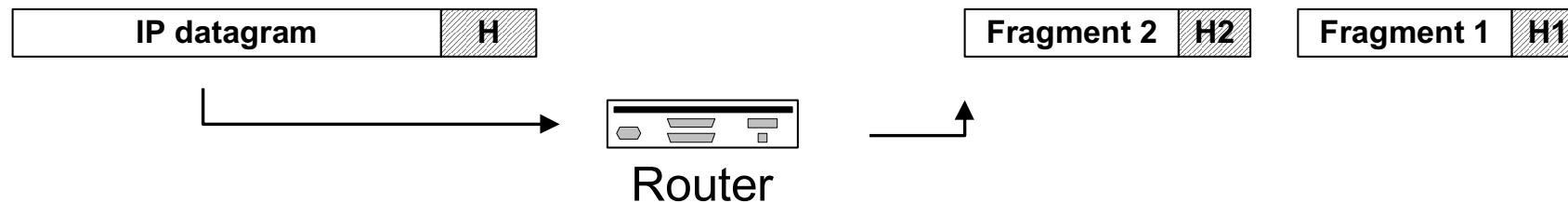


MTUs:      FDDI: 4352                      Ethernet: 1500

- **Fragmentation:**
  - IP router splits the datagram into several datagram
  - Fragments are reassembled at receiver

# Where is Fragmentation done?

- ▶ Fragmentation can be done at the sender or at intermediate routers
- ▶ The same datagram can be fragmented several times.
- ▶ Reassembly of original datagram is only done at destination hosts !!



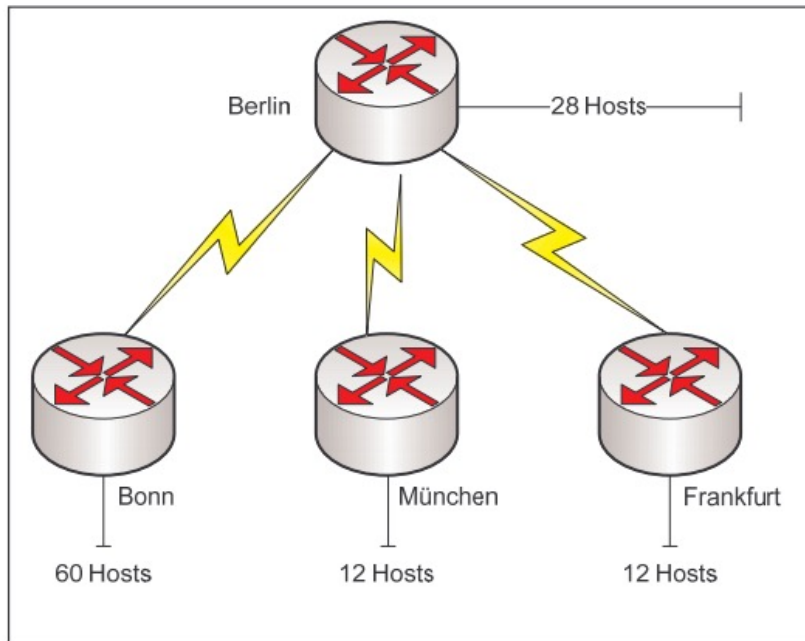
# What's involved in Fragmentation?

|                    |               |          |     |                         |        |        |
|--------------------|---------------|----------|-----|-------------------------|--------|--------|
| version            | header length | DS       | ECN | total length (in bytes) |        |        |
| Identification     |               |          |     | 0                       | D<br>F | M<br>F |
| Fragment offset    |               |          |     |                         |        |        |
| time-to-live (TTL) |               | protocol |     | header checksum         |        |        |

- ▶ The following fields in the IP header are involved:
  - ▶ **Identification** : When a datagram is fragmented, the identification is the same in all fragments
  - ▶ **Flags**
    - ▶ DF bit is set: Datagram cannot be fragmented and must be discarded if MTU is too small
    - ▶ MF bit set: This datagram is part of a fragment and an additional fragment follows this one
  - ▶ **Fragment offset**: Offset of the payload of the current fragment in the original datagram
  - ▶ **Total length**: Total length of the current fragment

# VLSM Subnetting

- ▶ Using classful subnetting wastes IP addresses
- ▶ The number of hosts involved in each subnet must of same size
- ▶ Create a number of subnet masks that suit our needs more efficiently than a classful subnetting scheme could



Example

